

Research Assignment 2: The Language of Business Analytics

1) FOUNDATIONAL CONCEPTS OF ANALYTICS

(1.1) Business Analytics

According to IBM.com

A Business Analytics refers to the statistical methods and computing technologies for processing, mining and visualizing data to uncover patterns, relationships and insights that enable better business decision-making.

Example:

- (1) Amazon (Product Recommendation): Amazon analyses user browsing behaviour, past purchases and items in the cart to provide personalized product recommendations. This data-driven approach is responsible for 35% of their total sales.

1.2 DATA ANALYTICS

Data analytics can optimize business performance by measuring and weighing raw data to gain information and draw conclusions from it (www.investopedia.com).

Example:

- (1) Real-time Pricing (Gig Economy)

Airbnb uses data to suggest competitive prices for hosts, accounting for location, season and local demand.

1.3 Business Intelligence (BI)

According to Business intelligence uncovers insights for making strategic decisions. Business intelligence tools

analyze historical and current data and present findings in intuitive visual format (www.microsoft.com).

1.4 DATA - DRIVEN DECISION MAKING

Data - driven decision making is an approach that emphasizes using data and analysis instead of intuition to inform business decision. By collecting, analyzing and interpreting data, organizations can make better decisions that more closely align with business goals and objectives. (www.ibm.com)

EXAMPLE:

- Financial Sector (Banking): Banks use data to reduce risk, manage fraud and target clients based on income and preferences

1.5 BIG DATA

Big data refers to massive, complex data sets that traditional data management systems cannot handle.

EXAMPLES

Entertainment: Content creation and recommendations like Netflix, Spotify.

1.6 Descriptive Analytics

Descriptive analytics is the process of using current and historical data to identify trends and

relationships (~~www~~ online.hbs.edu)

EXAMPLE: Software such as Microsoft Excel, or data visualization tools such as Google Charts and Tableau, can help parse data, identify trends and relationships between variables, and visually display information.

1.7 Diagnostic Analytics

Diagnostic analytics is a branch of business analytics that analyzes historical datasets to uncover root causes, patterns and relationships. It uses techniques such as data mining, correlation analysis, drill-down methods and statistical modeling.

EXAMPLE: Customer Support: Identifying that a high volume of support tickets is due to a new software bug in a specific feature.

1.8 Predictive Analytics

Predictive analytics is the use of data to predict future trends and events. The predictions could be for the near future - for instance predicting the malfunction of a piece of machinery later that day - or the more distant future such as predicting your company's cash flow for the upcoming year.

EXAMPLE

Energy & Utilities: Load Forecasting: Utility companies use weather forecasts, historical data and seasonal trends

to predict energy consumption.

1.9 Prescriptive Analytics

— To the process of using data to determine an optimal course of action. By considering all relevant factors, this type of analysis yield recommendation for next steps.

EXAMPLE : Retail Inventory and Pricing : Retailers analyze historical sales, seasonal trends and competitor price to set optimal stock levels and adjust prices automatically, ensuring high-demand items are available while reducing overstock cost.

1.10 Data Literacy

Data literacy is the ability to read, understand, use and communicate with data for better decision-making (www.ibm.com).

EXAMPLE : Operations and Logistics : Companies monitor project management dashboards to track KPIs and identify bottlenecks in real-time, moving away from relying solely on end-of-quarter reports.

2) DATA MEASURING AND REPORTING

2(11) Structured Data

Structured data is data that has a standardized format for efficient access by software and humans alike. Computers can effectively process structured data for insights due to its quantitative nature. For example a structured customer data table containing columns - name, address and phone number can provide insights like the total number of customers and the locality with the maximum number of customers.

(12) Unstructured Data

Unstructured data is information that does not have a predefined format.

Example: Social media post: Comments on Facebook contain unstructured text as well as hashtags.

13) Data Warehouse

Data warehouse is a type of data management system that is designed to enable and support business intelligence activities, especially analytics. It is intended to perform queries and analysis and often contain large amounts of historical data.

Examples: A ~~manufacturing~~ manufacturer utilized a data warehouse to unify scattered ERP and Production data, improving supply chain visibility and automating reports.

14) Data mining

It is the process of extracting and finding patterns in massive data sets involving methods at the intersection of machine learning, statistics and database systems.

Example : Hospital use data mining to analyze patient records, predict potential diseases and optimize treatment plans by finding correlations in vast medical datasets.

15) DATA VISUALIZATION

Data visualization is the representation of data through use of common graphics, such as charts, plots, infographics and even animation. These visual displays of information communicate complex data relationships and data-driven insights in a way that is easy to understand.

Example: Similar to analyzing communication a network graph can show the intensity of contact between two people, where lines get thicker or colors change based on the volume of messages or phone calls, mapping out the evolution of a connection.

16) Dashboard

A dashboard is a way of displaying various types of visual data in one place. (tableau.com).

Example is like the OKR dashboard which tracks high-level company goals to monitor progress on initiatives.

17) Metric

A metric is a calculated value that tracks performance for a business activity. (parameters.app).

EXAMPLE: Performance and Productivity: Average handling time is the time taken to resolve a customer service call.

18) Key Performance Indicator

Key performance indicators are the elements of an organization's plan that express the quantitative outcomes you seek and how you will measure success. In other words, it tells you what you want to achieve and ^{by} when.

EXAMPLE: Sales KPI: Sales growth, new sales from new customers, pipeline volume, average purchase value and win/loss ratio.

19) Benchmark

A benchmark is a powerful management tool that helps business compare their performance level against direct competitors, industry best practices

An example is by benchmarking your cost against industry average you can discover areas in which you're overspending then act to reduce overheads

20) SEGMENTATION

Is the process to separate a large group of prospective buyers into smaller groups for more focused marketing (Investopedia.com)

Example: Macdonald developed the best-dense segmentation strategy based on geography.

21)

(3) CUSTOMER & SALES KPI

21) Conversion Rate

Is the percentage of users who complete a desired action, such as clicking on an ad, installing an app ~~and~~ or making purchase

Example of Conversion rate is measuring the value of one currency against another. As the rate changes one country's money can become weaker or stronger against other currencies.

22) Customer Acquisition Cost

Is the cost related to acquiring a new customer.

Example a company spend 10000 ZAR on google ads campaign resulting in 10 new customers.

23) Customer Lifetime Value

— It helps business better understand their current customer base and who they should target to maximize profits.

An Example is grocery chain where a customer spend R50 per visit 26 times per year for seven years.

24) Churn Rate

The Churn rate is the percentage of customers or subscribers that discontinue their relationship with a business over a specific period, indicating the company's ability to retain clients and overall health.

An example of churn rate is a particularly.

Useful measurement in an industry like Telecommunications. This includes Cable or Satellite television providers, internet providers and telephone service providers. This is an industry in which most customers have multiple providers from which to choose and the cost of switching between them is not high.

25) Customer Retention Rate

Measures the percentage of customers who continue doing business with you.

An example is e-commerce like tortuga who addressed low purchase frequency for durable backpacks by doing product bundling, increasing repeat business.

26) Ner Promoter Score

Is a customer experience metric that helps organizations better understand customer loyalty and how happy customers are with their products and customer experience.

An example involves a single-question survey, to gauge customer loyalty. On a scale of 0-10, how likely are you to recommend our product to friends and family.

27) Average Order Value

It estimates the typical amount spent by a customer in each order commonly placed on a website or mobile app.

Example is paid ads might yield an AOV of X amount while organic traffic show AOV of Y amount. Prompting for example to adjust marketing efforts

28) Sales Funnel

A Sales funnel is a visual roadmap of the customer journey from initial awareness to final purchase.

An example of a sales funnel is a webinar funnel where a brand uses a free, live training session to educate prospects on a problem, culminating in a pitch for their software.

(4) MARKETING 4 CAMPAIGN ANALYTICS

29) Marketing Campaign

A marketing campaign is a set of strategic activities that promote a business goal or objectives.

An example is Nike "Just Do It" — instead of solely highlighting product features.

Nike focused on the mindset of determination using real, gritty stories of everyday athletes rather than just stars.

30) Post-campaign Analysis

A post-campaign or advertising report is a thorough analysis of the executed media plan. This allows for the evaluation of effectiveness of an advertising campaign. This process involves assessing all stages of the implementation of advertising materials and their impact on target audience.

Example is like Unilever and L'Oréal.

31) Incremental Revenue

Measure the net ^{new} revenue generated by a Campaign.

A Example: A food company launches a new flavor (Coke or new chocolate) to gain additional revenue from customers who might not have purchased the standard product, creating incremental income.

32) Return on Investment

Is a performance metric used to evaluate the efficiency or profitability of an investment.

Example: Home innovation: Spending

money to remodel a kitchen that increase a home selling price.

33) Return on Ad Spend

Is the amount of revenue earned for every dollar spent on a campaign.

An example is a auto shop runs a google display ad to promote special on brake replacement.

34) Click-Through Rate

Measures the percentage of ad viewers who click on an advertisement compare to the total number of ad impressions.

EXAMPLE: If 1000 people receive an marketing email and 50 click the link the CTR is 5%.

35) Bounce Rate

It represents the percentage of visitors who enter the site and then leave rather than continuing to view other pages within the same site.

EXAMPLE: By simplifying navigation and improving design, a retailer reduced their site bounce rate by 60% resulting in higher engagement.

36) A/B Testing

It consist of a randomized experiment that usually involves two variants (A+B) although the concept can be also extended to multiple variants of the same variable.

Examples: E-commerce stores often test displaying different price formats to see which encourage more purchases.

37) Lead Generation

In marketing lead generation refers to the process of attracting and capturing customer interest in a product or service.

Example: Offer a free trial.

38) Marketing Attribution

Is the process of identifying which marketing touchpoints or channels led a user to install an app.

Example: A customer sees an Instagram Ad searches on Google and eventually buys v.a Facebook.

Reference list

- 1) Hbm. Com.
- 2) www.investopedia.com
- 3) [povometrics.app](#)
- 4) www.tableau.com
- 5) www.geckco.com

